

IKA, Russian Federation

WMO: **300280**

Lat: **59.311N** Lon: **106.347E** Elev: **1153** StdP: **14.09** Time Zone: **8.00 (E08)** Period: **82-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**1**	**-53.9**	**-48.8**	**-57.9**	**0.2**	**-53.9**	**-53.1**	**0.2**	**-48.6**	**16.4**	**5.5**	**14.8**	**4.2**	**0.8**	**180**	**N/A**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**7**	**24.8**	**83.5**	**64.1**	**79.8**	**62.8**	**76.2**	**61.2**	**66.2**	**78.3**	**64.4**	**76.7**	**62.7**	**73.4**	**5.4**	**50**	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**62.0**	**86.8**	**70.4**	**60.1**	**80.9**	**68.4**	**58.1**	**75.2**	**66.9**	**31.7**	**78.0**	**30.2**	**76.8**	**28.9**	**73.7**	**77.2**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**16.2**	**14.1**	**12.1**	DB	**-59.3**	**89.0**	**3.1**	**2.5**	**-61.5**	**90.8**	**-63.4**	**92.3**	**-65.1**	**93.7**	**-67.4**	**95.5**	(4)
(5)				WB	**-59.3**	**69.9**	**3.1**	**2.9**	**-61.5**	**72.0**	**-63.4**	**73.7**	**-65.1**	**75.3**	**-67.4**	**77.4**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	22.5	-18.2	-10.7	7.5	25.7	40.9	57.1	60.5	55.1	41.5	22.7	-1.3	-13.5	(6)
(7)			DBStd	30.52	17.43	16.50	13.87	10.97	7.69	7.56	5.68	5.91	7.70	11.57	18.05	18.32	(7)
(8)			HDD50	10830	2114	1700	1319	730	298	23	1	21	273	845	1538	1968	(8)
(9)			HDD65	15568	2579	2120	1784	1180	748	250	161	309	706	1310	1988	2433	(9)
(10)			CDD50	777	0	0	0	0	16	238	326	179	18	0	0	0	(10)
(11)			CDD65	37	0	0	0	0	0	14	21	2	0	0	0	0	(11)
(12)			CDH74	1203	0	0	0	0	13	544	509	131	6	0	0	0	(12)
(13)			CDH80	287	0	0	0	0	0	148	116	23	0	0	0	0	(13)
(14)	Wind		WSAvg	4.3	3.1	3.5	4.8	5.7	5.7	4.8	4.1	3.9	4.0	4.5	4.0	3.6	(14)
(15)	Precipitation		PrecAvg	11.3	0.4	0.3	0.3	0.6	1.1	1.8	2.3	1.8	1.1	0.9	0.6	0.4	(15)
(16)			PrecMax	15.5	0.9	0.6	1.0	1.4	2.5	3.3	6.0	3.6	2.5	1.7	1.0	0.9	(16)
(17)			PrecMin	7.4	0.1	0.0	0.1	0.2	0.3	0.4	0.4	0.4	0.4	0.2	0.3	0.1	(17)
(18)			PrecStd	2.3	0.2	0.2	0.2	0.3	0.5	0.8	1.6	0.8	0.5	0.4	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		25.2	29.5	45.5	57.6	76.0	88.1	87.0	83.9	73.6	50.8	40.4	29.0	(19)
(20)			MCWB		23.2	26.2	36.6	43.5	54.7	65.4	65.4	64.2	58.4	41.2	37.5	26.4	(20)
(21)		2%	DB		18.7	25.1	39.2	52.7	68.4	84.4	83.0	77.5	66.0	45.2	33.4	24.1	(21)
(22)			MCWB		17.0	22.6	33.0	41.6	51.7	63.7	64.7	61.8	53.5	39.0	31.2	22.4	(22)
(23)		5%	DB		13.8	20.2	33.9	47.7	63.3	80.6	79.7	73.7	59.9	40.6	29.0	19.8	(23)
(24)			MCWB		12.5	18.2	29.4	38.1	49.5	62.4	63.9	60.3	50.7	36.0	26.7	18.5	(24)
(25)		10%	DB		8.6	14.9	29.6	43.0	57.9	76.8	76.1	69.5	55.5	36.8	25.3	14.8	(25)
(26)			MCWB		7.5	13.3	25.9	35.4	46.2	60.5	62.1	58.5	48.3	33.6	23.4	13.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		23.2	26.5	37.6	44.9	56.8	67.9	69.0	66.7	60.1	42.5	37.8	27.1	(27)
(28)			MCDB		24.6	29.1	43.4	55.3	72.9	81.3	79.7	75.9	70.9	48.5	39.9	29.1	(28)
(29)		2%	WB		17.0	23.0	33.3	41.8	52.8	65.6	66.7	64.0	55.7	39.4	31.4	22.6	(29)
(30)			MCDB		18.6	25.3	38.7	51.7	66.5	80.4	78.0	74.2	63.6	44.6	33.2	24.2	(30)
(31)		5%	WB		12.5	18.2	29.7	38.9	50.0	63.7	65.1	62.2	52.1	36.0	26.8	18.5	(31)
(32)			MCDB		13.8	20.3	33.8	46.3	61.7	77.6	76.6	71.0	58.0	39.5	28.7	19.7	(32)
(33)		10%	WB		7.5	13.4	26.2	36.1	47.3	61.7	63.4	59.9	48.9	33.9	23.4	13.5	(33)
(34)			MCDB		8.7	15.0	29.7	42.6	56.9	74.7	74.0	67.6	54.6	36.8	25.4	14.6	(34)
(35)	Mean Daily Temperature Range		MDBR		21.1	28.2	31.5	24.8	22.8	27.1	24.8	24.7	21.5	17.1	19.2	18.8	(35)
(36)		5% DB	MCDBR		23.4	25.9	30.5	29.3	33.4	37.1	33.6	34.1	31.1	21.8	15.1	18.7	(36)
(37)			MCWBR		22.4	23.5	24.6	19.4	18.8	18.3	16.8	19.0	19.4	15.8	14.5	17.9	(37)
(38)		5% WB	MCDBR		23.5	24.8	29.2	26.2	31.3	32.0	28.5	27.8	28.5	19.7	15.0	19.1	(38)
(39)			MCWBR		22.5	22.5	23.7	17.8	18.5	16.6	15.7	16.7	18.7	15.1	14.3	18.4	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.256	0.292	0.314	0.346	0.376	0.409	0.479	0.425	0.339	0.308	0.258	0.233	(40)
(41)		taud			1.976	2.011	2.053	2.115	2.283	2.292	2.071	2.196	2.438	2.298	2.028	1.925	(41)
(42)		Ebn at Noon			160	219	255	269	270	261	237	242	252	219	158	122	(42)
(43)		Edh at Noon			17	28	36	41	37	38	46	38	25	21	15	12	(43)
(44)	All-Sky Solar Radiation	RadAvg			176	481	971	1461	1642	1817	1658	1307	804	387	202	102	(44)
(45)		RadStd			12	31	62	85	105	137	143	95	77	43	21	10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(46)
(47)	Regional (3 neighbors)	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(47)

Nomenclature: See separate page